

OSLAB SKILL-7 PROGRAMS & OUTPUTS

Date: 04-10-2026

Program-1 Code & Output:

```
sainath@Sainath:~/OSLAB$ mkdir skill7
sainath@Sainath:~/OSLAB$ cd skill7
sainath@Sainath:~/OSLAB/skill7$ gedit program1.c
sainath@Sainath:~/OSLAB/skill7$ gcc program1.c -o program1
sainath@Sainath:~/OSLAB/skill7$ cat program1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int status;
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Child: Running...\n");
        sleep(2);
        printf("Child: Finished\n");
        return 5;
    }

    printf("Parent: Waiting for child...\n");

    waitpid(pid, &status, 0);

    if (WIFEXITED(status))
        printf("Parent: Child exited with status %d\n",
               WEXITSTATUS(status));

    return 0;
}
sainath@Sainath:~/OSLAB/skill7$ ./program1
Parent: Waiting for child...
Child: Running...
Child: Finished
Parent: Child exited with status 5
```

Program-2 Code & Output:

```
sainath@Sainath:~/OSLAB/skill7$ gedit program2.c
sainath@Sainath:~/OSLAB/skill7$ gcc program2.c -o program2
sainath@Sainath:~/OSLAB/skill7$ cat program2.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

int main() {
    char *path = getenv("PATH");
    char *copy, *dir;
    char file[200];
    char command[50];

    printf("Enter command: ");
    scanf("%49s", command);

    copy = strdup(path);
    dir = strtok(copy, ":");

    while (dir != NULL) {
        snprintf(file, sizeof(file), "%s/%s", dir, command);

        if (access(file, X_OK) == 0) {
            printf("Executable found: %s\n", file);
            free(copy);
            return 0;
        }

        dir = strtok(NULL, ":");
    }

    printf("Command not found: %s\n", command);

    free(copy);
    return 0;
}
sainath@Sainath:~/OSLAB/skill7$ ./program2
Enter command: ls
Executable found: /usr/bin/ls
```

History:

```
642 cd OSLAB
643 ls
644 mkdir skill7
645 cd skill7
646 gedit program1.c
647 gcc program1.c -o program1
648 cat program1.c
649 ./program1
650 gedit program2.c
651 gcc program2.c -o program2
652 cat program2.c
653 ./program2
654 history
```